

Microsoft Azure Fundamentals (AZ-900) — Section 3 Lab

Azure Architecture: Resource Groups, Regions, Tagging & Governance

Lab Provider: Dion Training | **Estimated Duration:** 15 minutes

Platform: Microsoft Azure (Free Account) | **Tools Used:** Azure Portal · Resource Groups · Storage Accounts · Cost Management · ARM · IAM **Completed by** **OLUWASEUN OSUNSOLA** [View LinkedIn](#)

Lab Overview

Continuing as IT consultant for **CloudFirst Retail**, this lab focused on establishing a proper organizational structure in Azure before any production workloads are deployed. The goal was to demonstrate how Azure's resource hierarchy works, how region selection impacts deployments, and how a tagging strategy enables accurate cost tracking — all core concepts for AZ-900 and real-world cloud governance.

Learning Objectives

- Create and configure Azure Resource Groups with proper naming conventions
 - Deploy resources across different regions within the same Resource Group
 - Implement a tagging strategy and observe that tags do **not** auto-inherit to resources
 - Navigate the Azure organizational hierarchy: Subscriptions → Resource Groups → Resources
 - View ARM deployment history to understand Infrastructure as Code (IaC)
 - Explore RBAC role assignment and policy inheritance at the Resource Group level
-

Prerequisites

- Active Azure free account
- Access to the Azure Portal
- 15 minutes of focused time

Azure Services Used: Resource Groups · Storage Accounts · Azure Portal (Cost Management, Subscriptions, Deployments, Activity Log)

Step-by-Step Walkthrough

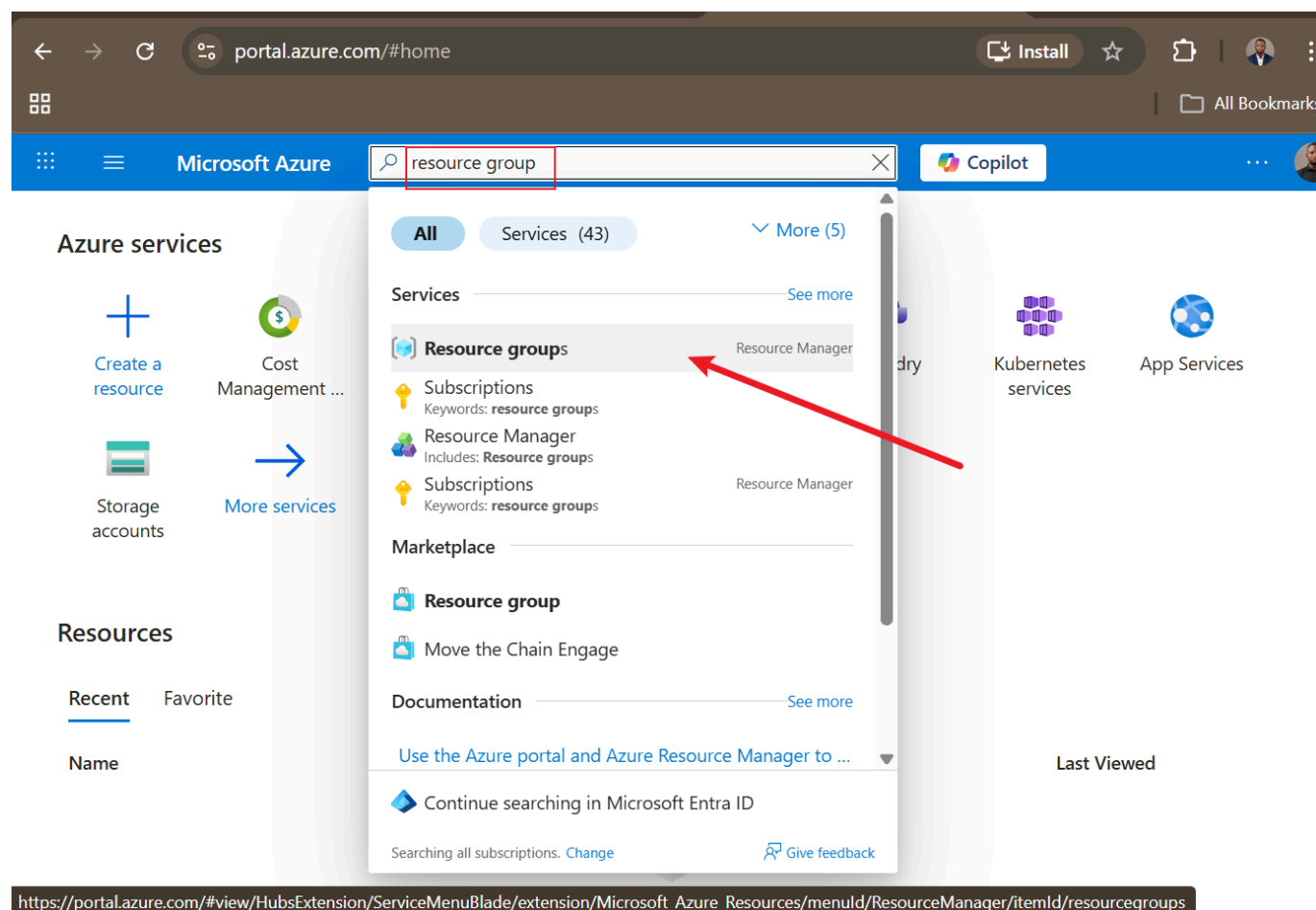
Step 1 — Sign In to the Azure Portal

Signed into the Azure Portal at <https://portal.azure.com>. This is the entry point for all Azure resource management.

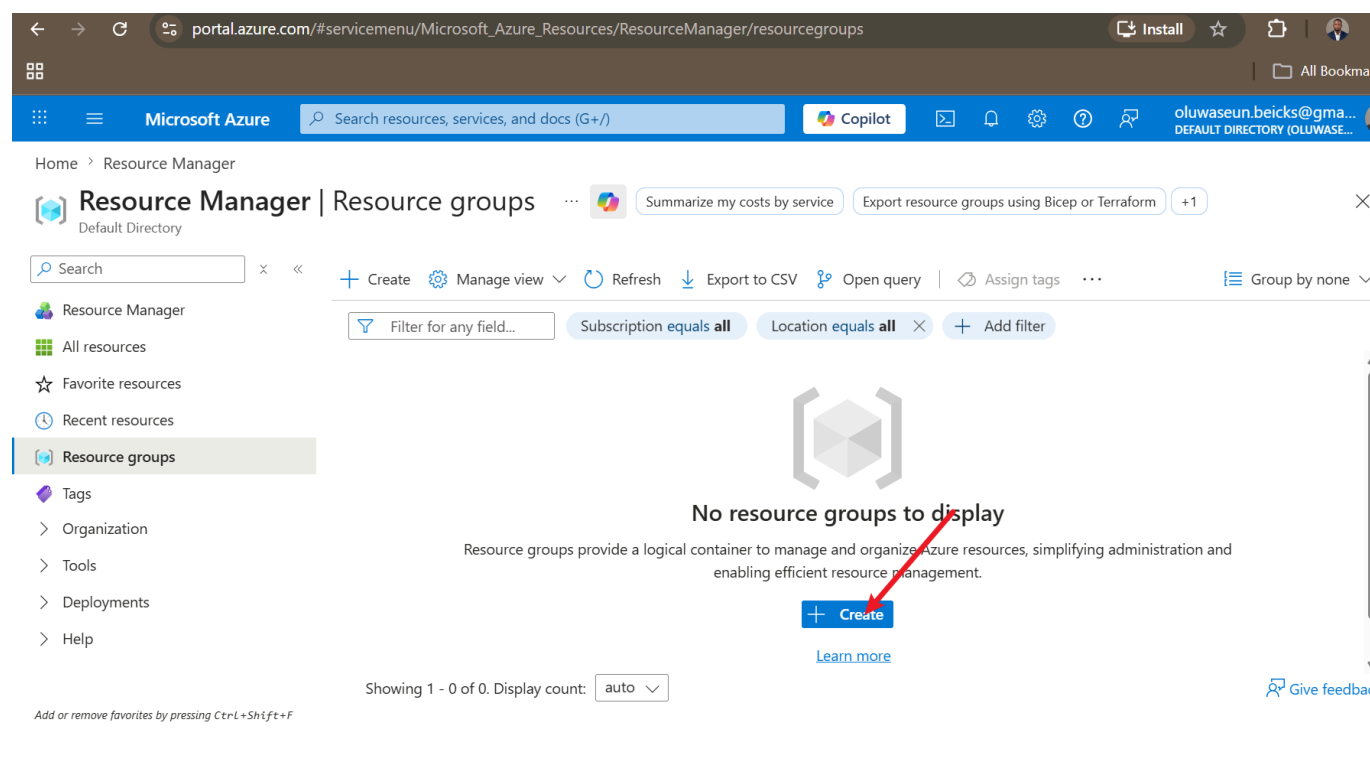
 Sign into Azure Portal

Step 2 — Navigate to Resource Groups

Searched for **Resource groups** in the top search bar and navigated to the Resource Groups page. Resource Groups are the logical containers that organise every Azure resource — this is where governance, access control, and cost tracking begin.



Clicked **+ Create** to begin creating a new Resource Group.



Step 3 — Create the CloudFirst Retail Resource Group

Configured the Resource Group with the following settings, following Azure naming best practices (**rg-[application]-[environment]-[region]**):

- **Subscription:** Azure Free Account
- **Resource group name:** **rg-cloudfirst-test-eastus**
- **Region:** East US

portal.azure.com/#view/HubsExtension/ResourceGroupCreate.ReactView/_provisioningContext~/7B"telemetryId"%3A"600e9949-a7d...

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > Resource Manager | Resource groups

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * ⓘ Azure for Students

Resource group name * ⓘ rg-cloudfirst-test-eastus

Region * ⓘ (US) East US

Previous Next Review + create

On the **Tags** tab, added three organisational tags to the Resource Group:

Name	Value
Environment	Test
Project	CloudFirst-Retail
Department	IT-Consulting

portal.azure.com/#view/HubsExtension/ResourceGroupCreate.ReactView/_provisioningContext~/~/%7B%22telemetryId%3A%22600e9949-a7d...

Microsoft Azure Search resources, services, and docs (G+)

Home > Resource Manager | Resource groups

Create a resource group

Basics **Tags** Review + create

Apply tags to your Azure resources to logically organize them by categories. A tag consists of a key (name) and a value. Tag names are case-insensitive and tag values are case-sensitive. [Learn more](#)

Name	Value	Resource
Environment	: Test	Resource group
Project	: CloudFirst-Retail	Resource group
Department	: IT-Consulting	Resource group
		Resource group

Previous Next **Review + create**

Reviewed all settings and clicked **Create**.

portal.azure.com/#view/HubsExtension/ResourceGroupCreate.ReactView/_provisioningContext~/~/%7B%22telemetryId%3A%22600e9949-a7d...

Microsoft Azure Search resources, services, and docs (G+)

Home > Resource Manager | Resource groups

Create a resource group

Basics Tags **Review + create**

Basics

Subscription Azure for Students

Resource group name rg-cloudfirst-test-eastus

Region East US

Tags

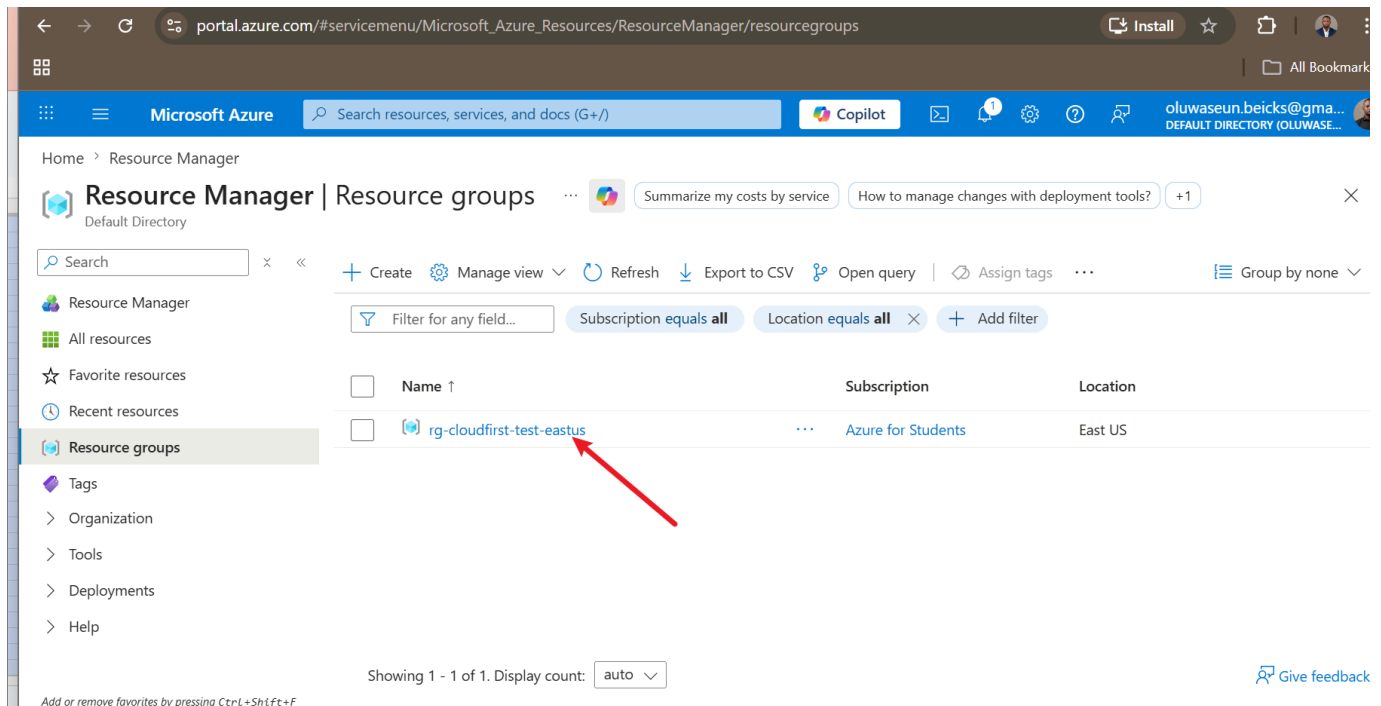
Environment Test

Project CloudFirst-Retail

Department IT-Consulting

Previous Next **Create**

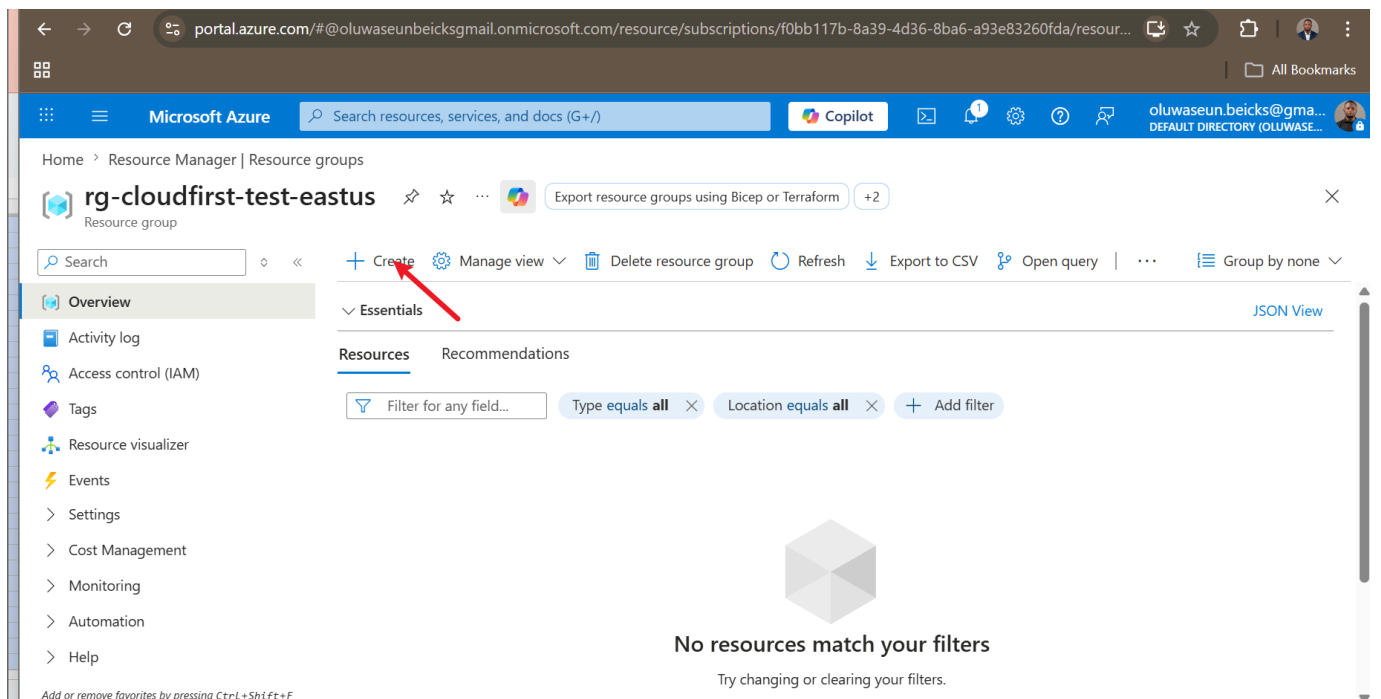
Resource Group **rg-cloudfirst-test-eastus** was successfully created. Clicked on it to enter the RG dashboard.



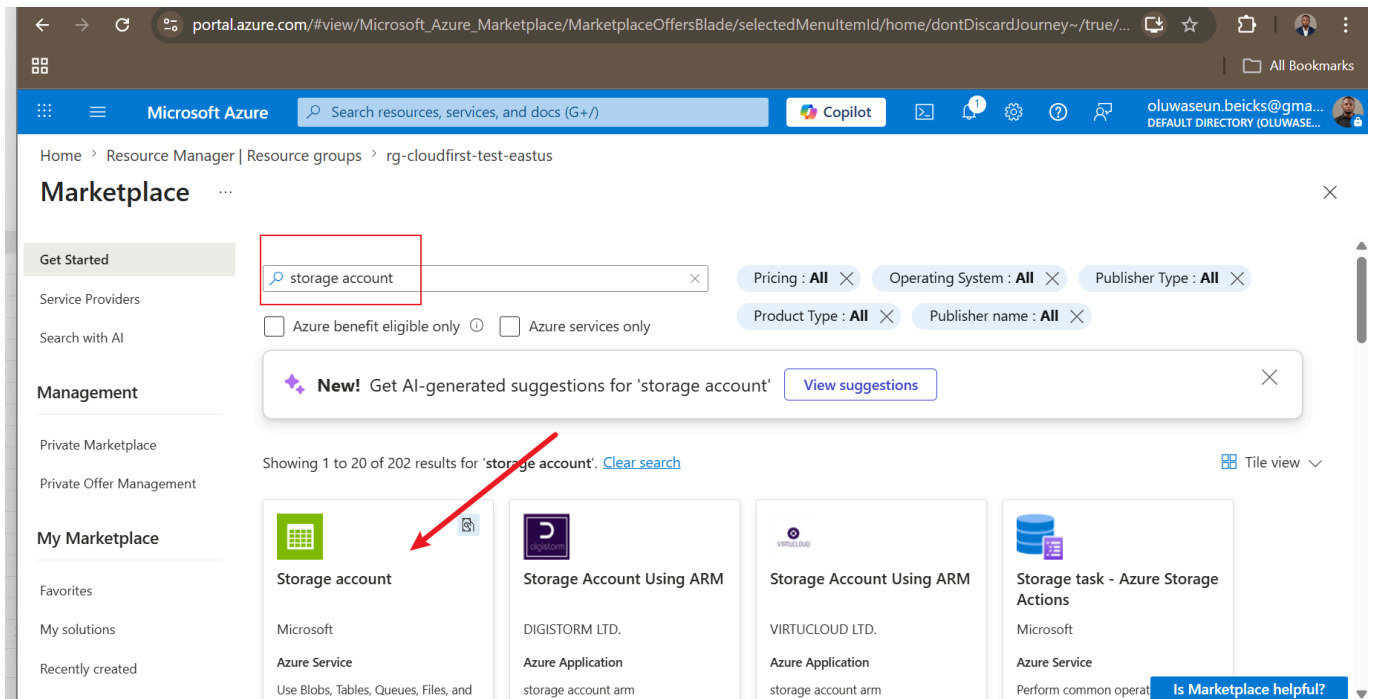
Why This Matters: The RG region only determines where its **metadata** is stored — not where the resources inside must live. This matters for compliance and data residency requirements.

Step 4 — Deploy a Storage Account in a Different Region

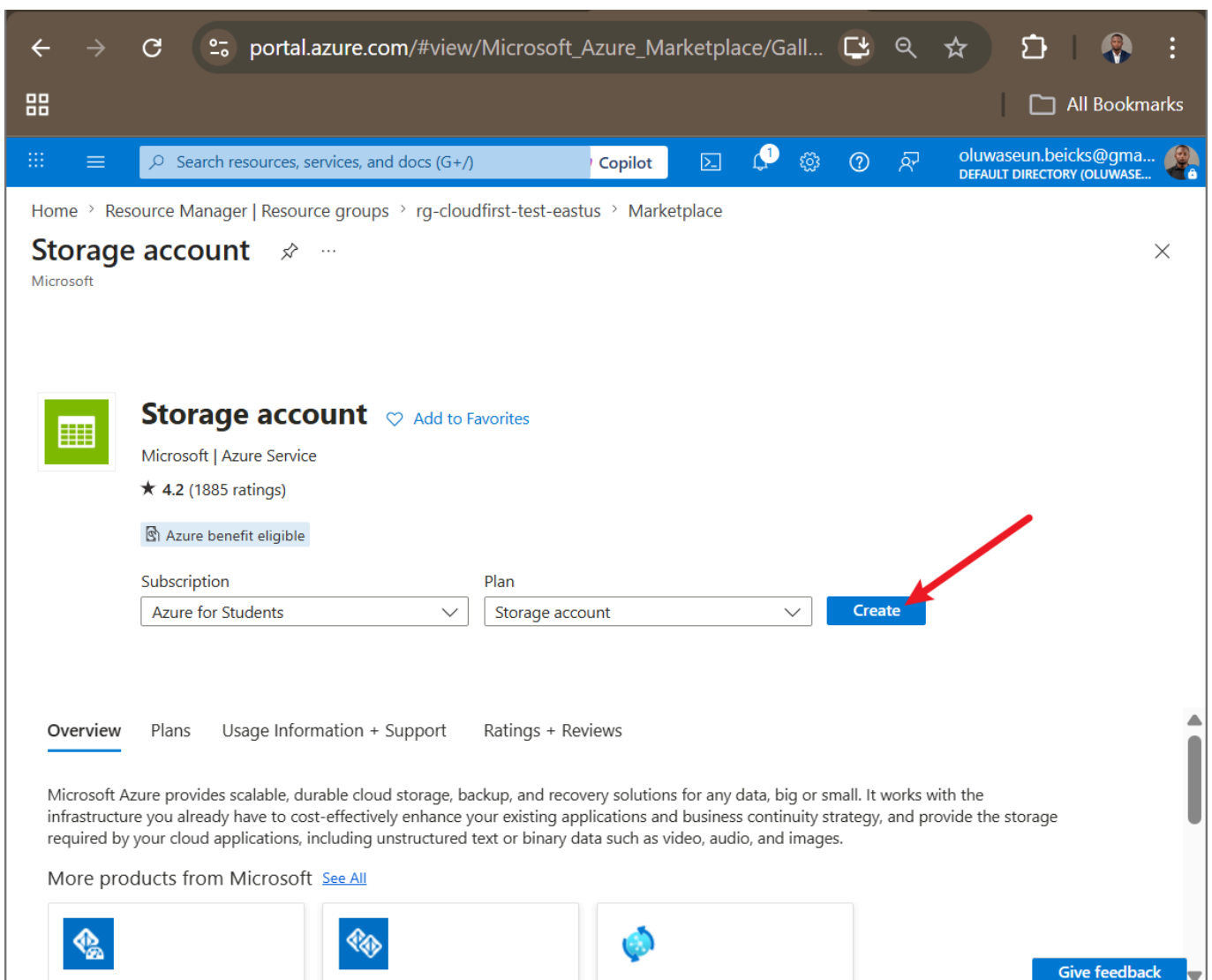
From inside the Resource Group, clicked + **Create** to deploy a resource.



Searched for **Storage account** and selected it from the results.



On the Storage Account creation page, confirmed the subscription and plan.



Configured the Storage Account with intentionally different region from the RG to demonstrate that Resource Groups are logical, not physical, boundaries:

- **Subscription:** Pre-selected
- **Resource group:** **rg-cloudfirst-test-eastus** (pre-selected)
- **Storage account name:** **cfretailtest234**
- **Region:** South Africa North (*different from the RG's East US region*)
- **Performance:** Standard
- **Redundancy:** Locally-redundant storage (LRS)

portal.azure.com/#create/Microsoft.StorageAccount-ARM

Home > Resource Manager | Resource groups > rg-cloudfirst-test-eastus > Marketplace > Storage account

Create a storage account

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *

Resource group *
[Create new](#)

Instance details

Storage account name *

Region *
[Deploy to an Azure Extended Zone](#)

Preferred storage type

This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. [Learn more](#)

Performance * ☒ **Standard:** Recommended for most scenarios (general-purpose v2 account)
☐ **Premium:** Recommended for scenarios that require low latency.

[Previous](#) [Next](#) [Review + create](#) [Give feedback](#)

Clicked **Create** to deploy.

portal.azure.com/#create/Microsoft.StorageAccount-ARM

Home > Resource Manager | Resource groups > rg-cloudfirst-test-eastus > Marketplace > Storage account

Create a storage account

[View automation template](#)

Basics

Subscription	Azure for Students
Resource group	rg-cloudfirst-test-eastus
Location	South Africa North
Storage account name	cfretailtest234
Preferred storage type	
Performance	Standard
Replication	Locally-redundant storage (LRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled

[Previous](#) [Next](#) [Create](#) [Give feedback](#)

Deployment completed successfully. Clicked **Go to resource**.

portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2F0bb117b-8a39-4d36-8ba6-a93...

Microsoft Azure

Home

cfretailtest234_1777346612817 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name: cfretailtest234_1777346612817
Subscription: Azure for Students
Resource group: rg-cloudfirst-test-eastus

Start time: 4/28/2026, 4:25:40 AM
Correlation ID: a98365a4-0a3f-42ea-8c82-477bfa063752

Deployment details

Next steps

Go to resource

Give feedback

Tell us about your experience with deployment

Cost Management

Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >

Free Microsoft tutorials

Start learning today >

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
Find an Azure expert >

Step 5 — Observe Cross-Region Deployment

In the storage account **Essentials** section, confirmed that the resource location shows **South Africa North** — even though the parent Resource Group is in **East US**. This proves that Resource Groups are purely logical containers with no physical boundary restrictions.

portal.azure.com/#@oluwaseunbeicksgmail.onmicrosoft.com/resource/subscriptions/f0bb117b-8a39-4d36-8ba6-a93e83260fda/resou...

Microsoft Azure

Home > cfretailtest234_1777346612817 | Overview

cfretailtest234

Create an LCM rule for my storage account Analyze alerts for this storage account Improve storage account data protection

Search

Upload Open in Explorer Delete Move Refresh Open in mobile CLI / PS Feedback

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Security + networking

Data management

Settings

Annotations

Essentials

Resource group (move) : rg-cloudfirst-test-eastus

Location : southafricanorth

Subscription (move) : Azure for Students

Subscription ID : f0bb117b-8a39-4d36-8ba6-a93e83260fda

Disk state : Available

Tags (edit) : Add tags

Performance : Standard

Replication : Locally-redundant storage (LRS)

Account kind : StorageV2 (general purpose v2)

Provisioning state : Succeeded

Created : 4/28/2026, 4:25:41 AM

Properties Monitoring Capabilities (7) Recommendations (0) Tutorials Tools + SDKs

Blob service

Hierarchical namespace : Disabled

Default access tier : Hot

Blob anonymous access : Disabled

Blob soft delete : Enabled (7 days)

Container soft delete : Enabled (7 days)

Versioning : Disabled

Change feed : Disabled

Security

Require secure transfer for REST API operations : Enabled

Storage account key access : Enabled

Minimum TLS version : Version 1.2

Infrastructure encryption : Disabled

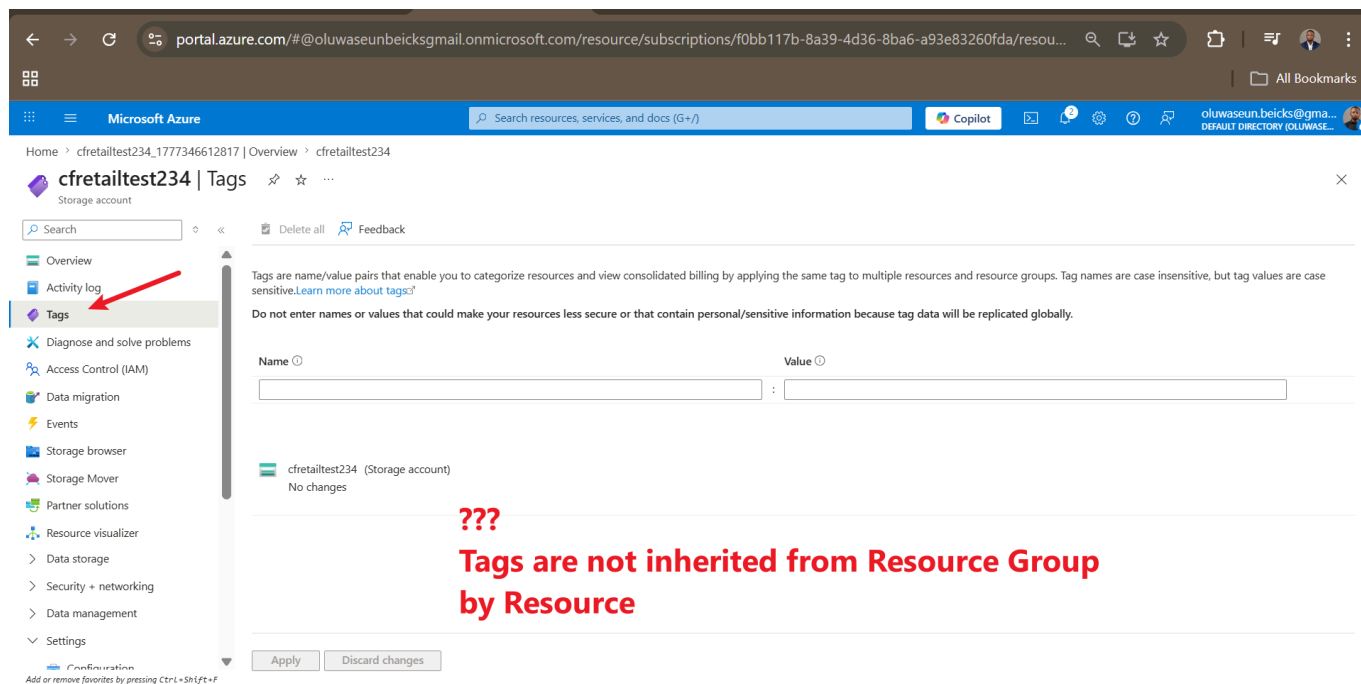
Networking

Durable network access : Enabled

Real-World Application: CloudFirst Retail could have development resources in South Africa, production infrastructure in East US, and manage them all under the same Resource Group for unified governance.

Step 6 — Observe Tag Non-Inheritance (Critical AZ-900 Concept)

Clicked **Tags** under Settings on the storage account. The Tags page was completely **empty** — even though the parent Resource Group had three tags applied.



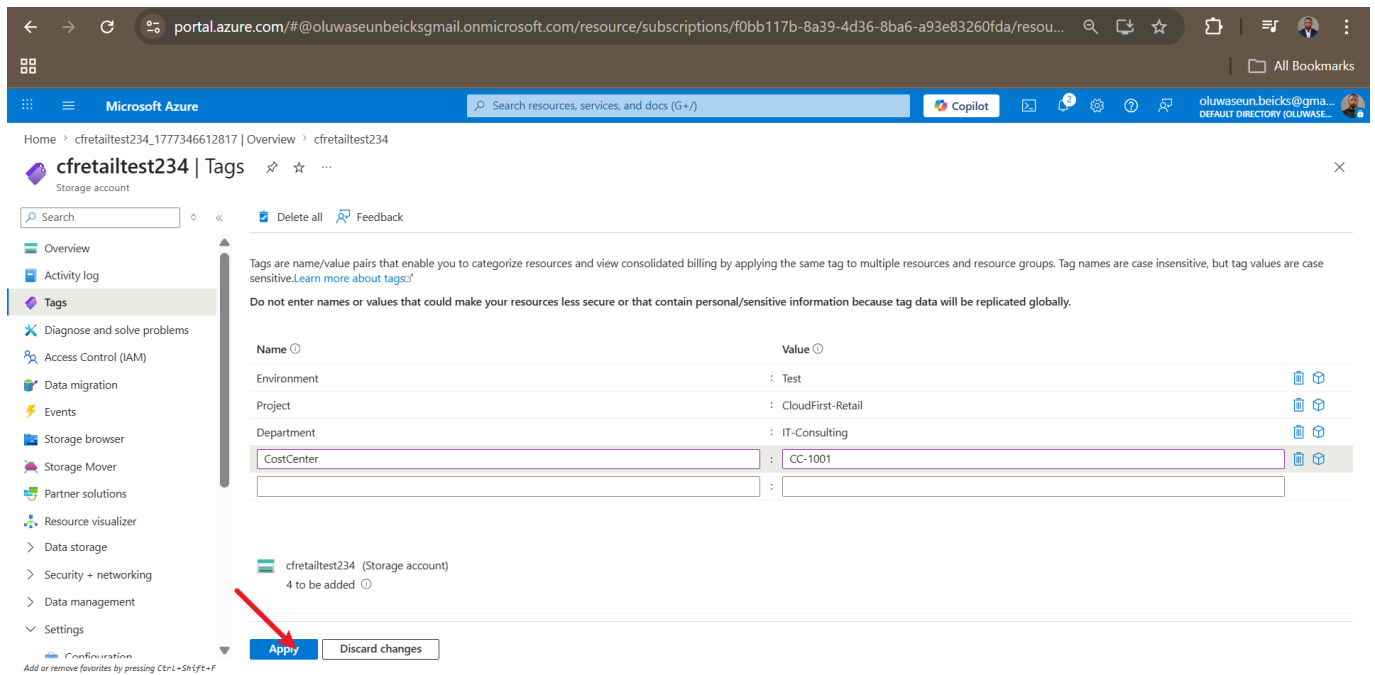
Key Concept — Exam Trap: Tags applied to a Resource Group do **NOT** automatically flow down to resources inside. This is one of the most commonly tested and most commonly misunderstood concepts in AZ-900 and in production environments.

Step 7 — Apply Tags Directly to the Storage Account

Manually applied four tags directly to the storage account to enable accurate cost tracking and organisational management:

Name	Value
Environment	Test
Project	CloudFirst-Retail
Department	IT-Consulting
CostCenter	CC-1001

Clicked **Apply** to save.



Home > cfretailtest234_1777346612817 | Overview > cfretailtest234

cfretailtest234 | Tags

Storage account

Search

Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Storage Mover
Partner solutions
Resource visualizer
Data storage
Security + networking
Data management
Settings

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. Tag names are case insensitive, but tag values are case sensitive. [Learn more about tags.](#)

Do not enter names or values that could make your resources less secure or that contain personal/sensitive information because tag data will be replicated globally.

Name	Value
Environment	Test
Project	CloudFirst-Retail
Department	IT-Consulting
CostCenter	CC-1001

cfretailtest234 (Storage account)
4 to be added

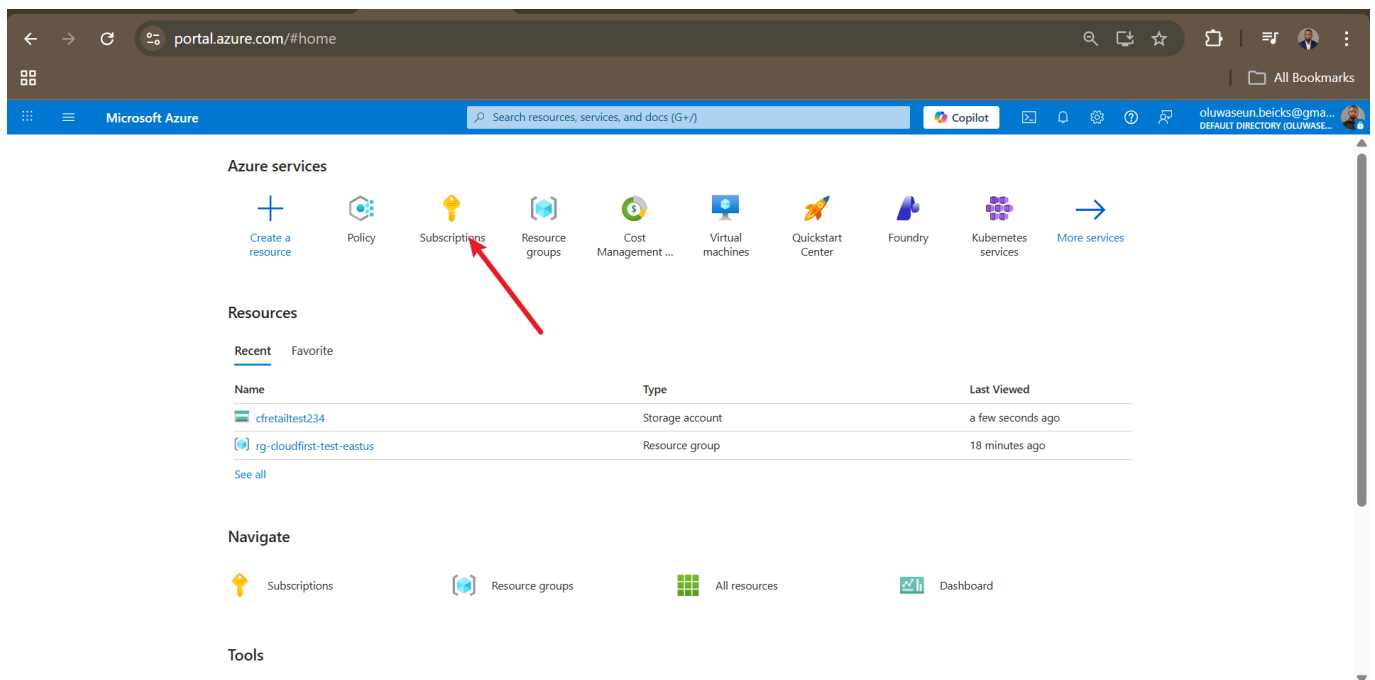
[Apply](#) [Discard changes](#)

Add or remove favorites by pressing Ctrl+Shift+F

Why This Matters: These tags now appear in billing reports and cost analysis dashboards. In production, Azure Policy is used to automatically enforce tagging requirements — blocking deployments that don't include required tags like **Project** and **CostCenter**.

Step 8 — View the Azure Resource Hierarchy

Navigated back home and searched for **Subscriptions** to trace the full Azure organisational hierarchy in action.



portal.azure.com/#home

Azure services

Create a resource | Policy | **Subscriptions** | Resource groups | Cost Management | Virtual machines | Quickstart Center | Foundry | Kubernetes services | More services

Resources

Recent | Favorite

Name	Type	Last Viewed
cfretailtest234	Storage account	a few seconds ago
rg-cloudfirst-test-eastus	Resource group	18 minutes ago

[See all](#)

Navigate

Subscriptions | Resource groups | All resources | Dashboard

Tools

Selected the active subscription.

portal.azure.com/#view/Microsoft_Azure_Billing/SubscriptionsBladeV2

Microsoft Azure

Subscriptions

Default Directory (oluwaseunbeicks@gmail.onmicrosoft.com)

Global administrators can manage all subscriptions in this list by updating their policy setting [here](#).

View list of subscriptions for which you have role-based access control (RBAC) permissions to manage Azure resources. To view subscriptions for which you have billing access, [click here](#).

Showing subscriptions in Default Directory directory. Don't see a subscription? [Switch directories](#)

Search for any field...

Subscriptions: Filtered (1 of 1) My role == all Status == all Add filter

Subscription name ↑↓	Subscription ID ↑↓	My role ↑↓	Current cost	Secure Score ↑↓	Parent management group ↑↓	Status ↑↓
Azure for Students	f0bb117b-8a39-4d36-8ba6-a93e83260fda	Owner	0.00	-	Tenant Root Group	Active

Inside the subscription dashboard, clicked on `rg-cloudfirst-test-eastus` under Resource Groups to confirm the storage account `cfretailtest234` is listed — visually confirming the hierarchy: **Subscription** → **Resource Group** → **Resource**.

portal.azure.com/#@oluwaseunbeicks@gmail.onmicrosoft.com/resource/subscriptions/f0bb117b-8a39-4d36-8ba6-a93e83260fda/resou...

Microsoft Azure

Subscriptions > Azure for Students | Resource groups

Azure for Students | Resource groups

rg-cloudfirst-test-eastus

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Showing 1 - 1 of 1. Display count: auto

rg-cloudfirst-test-eastus

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Showing 1 - 1 of 1. Display count: auto

Resources

Filter for any field...

Type equals all Location equals all Add filter

Name ↑	Type	Location
cfretailtest234	Storage account	South Africa North

Governance Insight: RBAC permissions and Azure Policies applied at the Subscription or Resource Group level **do** inherit downward to all resources within — unlike tags. This is how enterprises govern thousands of resources with minimal administrative effort.

Step 9 — Explore Cost Management with Tags

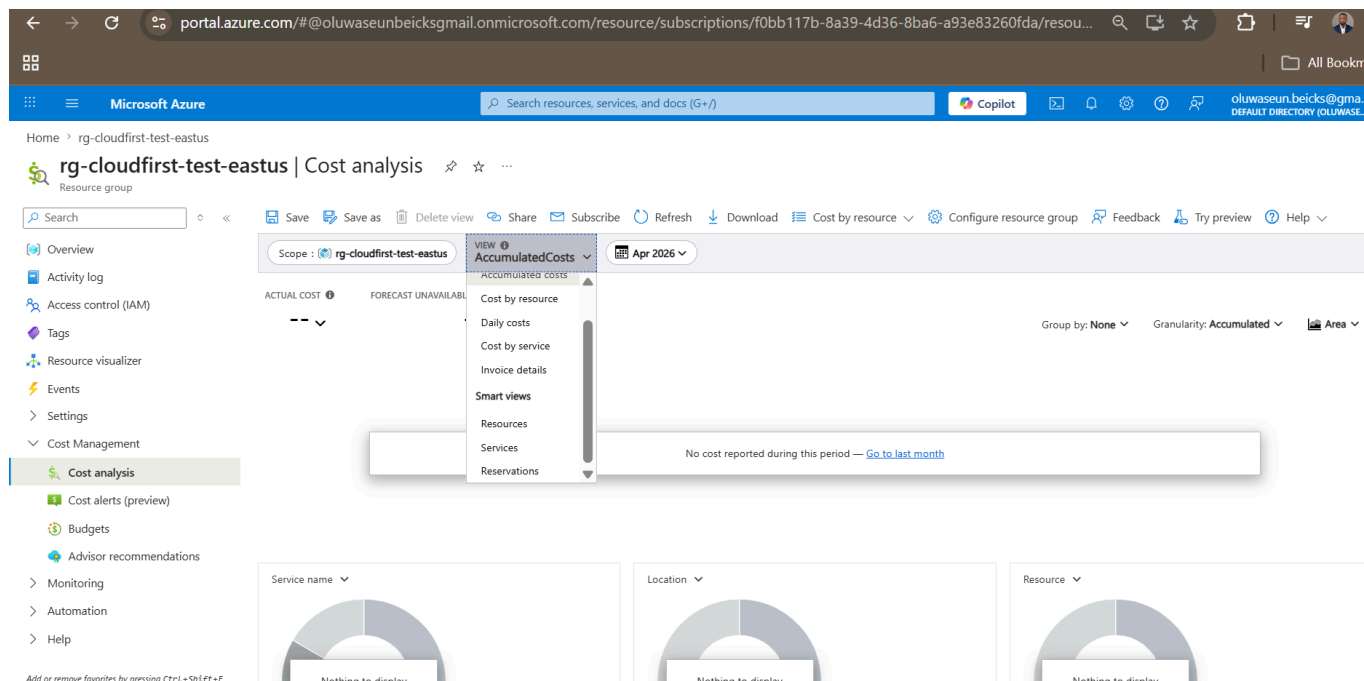
Inside the Resource Group, navigated to **Cost Management** → **Cost Analysis**.

The screenshot shows the Azure portal interface for the resource group 'rg-cloudfirst-test-eastus'. The left-hand navigation pane is visible, with 'Cost Management' highlighted. A red box and arrow point to the 'Cost analysis' link under 'Cost Management'. The main content area shows the 'Resources' tab with a table listing resources. The table has columns for Name, Type, and Location. One resource is listed: 'cfretailtest234' (Storage account) in 'South Africa North'.

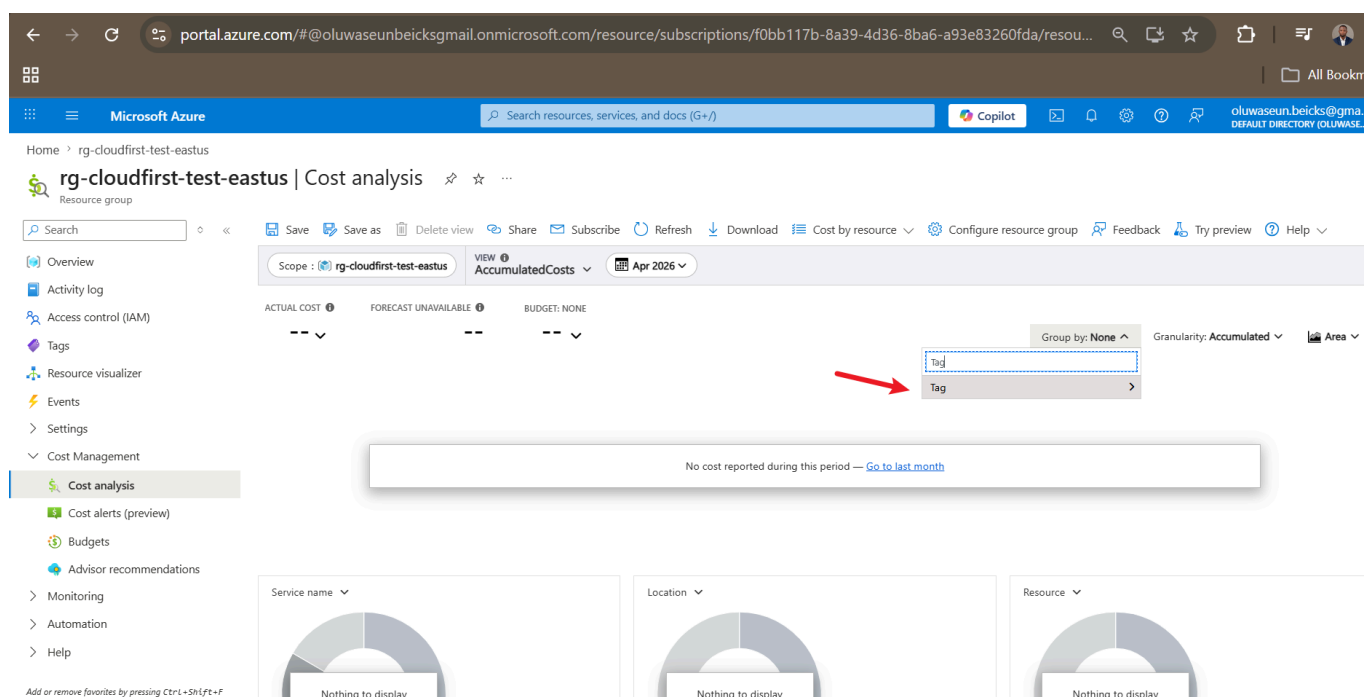
The dashboard showed no data — expected behaviour for newly created resources. Cost data typically takes **8–24 hours** to appear after resource creation.

The screenshot shows the 'Cost analysis' dashboard for the resource group 'rg-cloudfirst-test-eastus'. The dashboard displays a message: "No cost reported during this period — [Go to last month](#)". Below the message are three donut charts for 'Service name', 'Location', and 'Resource', all showing "Nothing to display". The dashboard also includes a 'VIEW' dropdown set to 'Accumulated Costs' and a date selector set to 'Apr 2026'.

Clicked **View** to explore the available cost breakdown filter options.



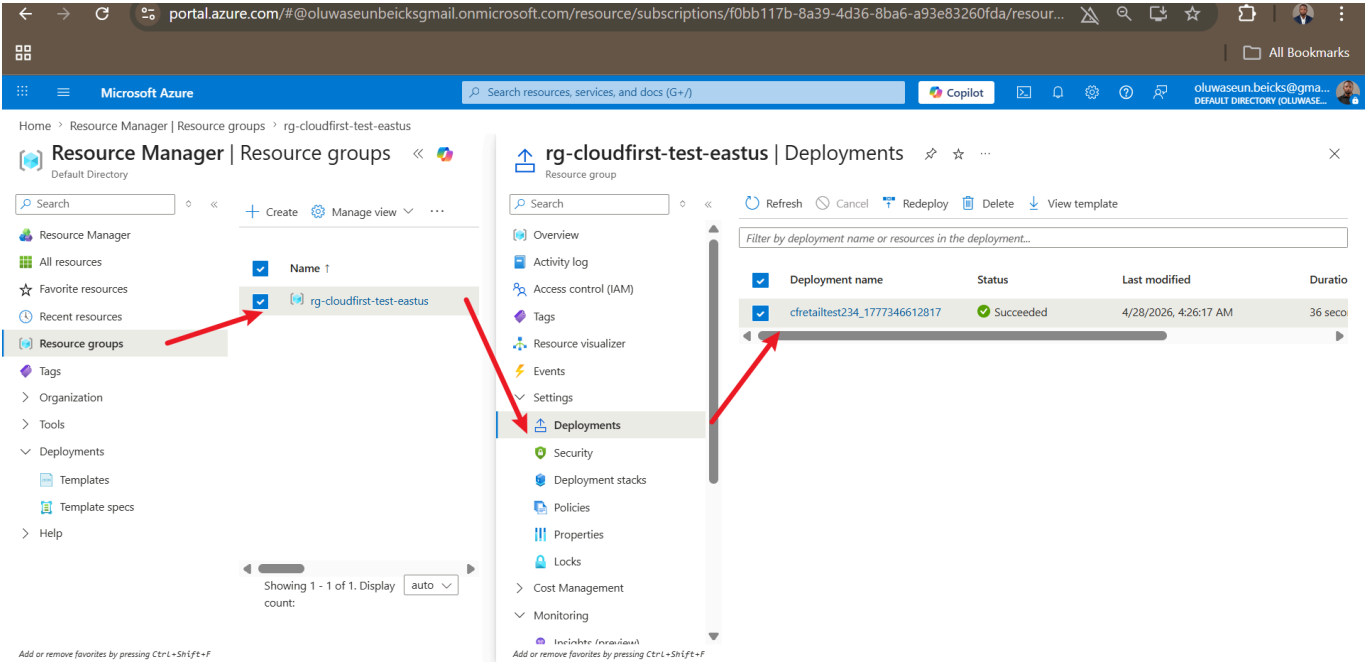
Noted that costs can be grouped and filtered by tags — enabling per-project, per-department, and per-environment cost attribution once data becomes available.



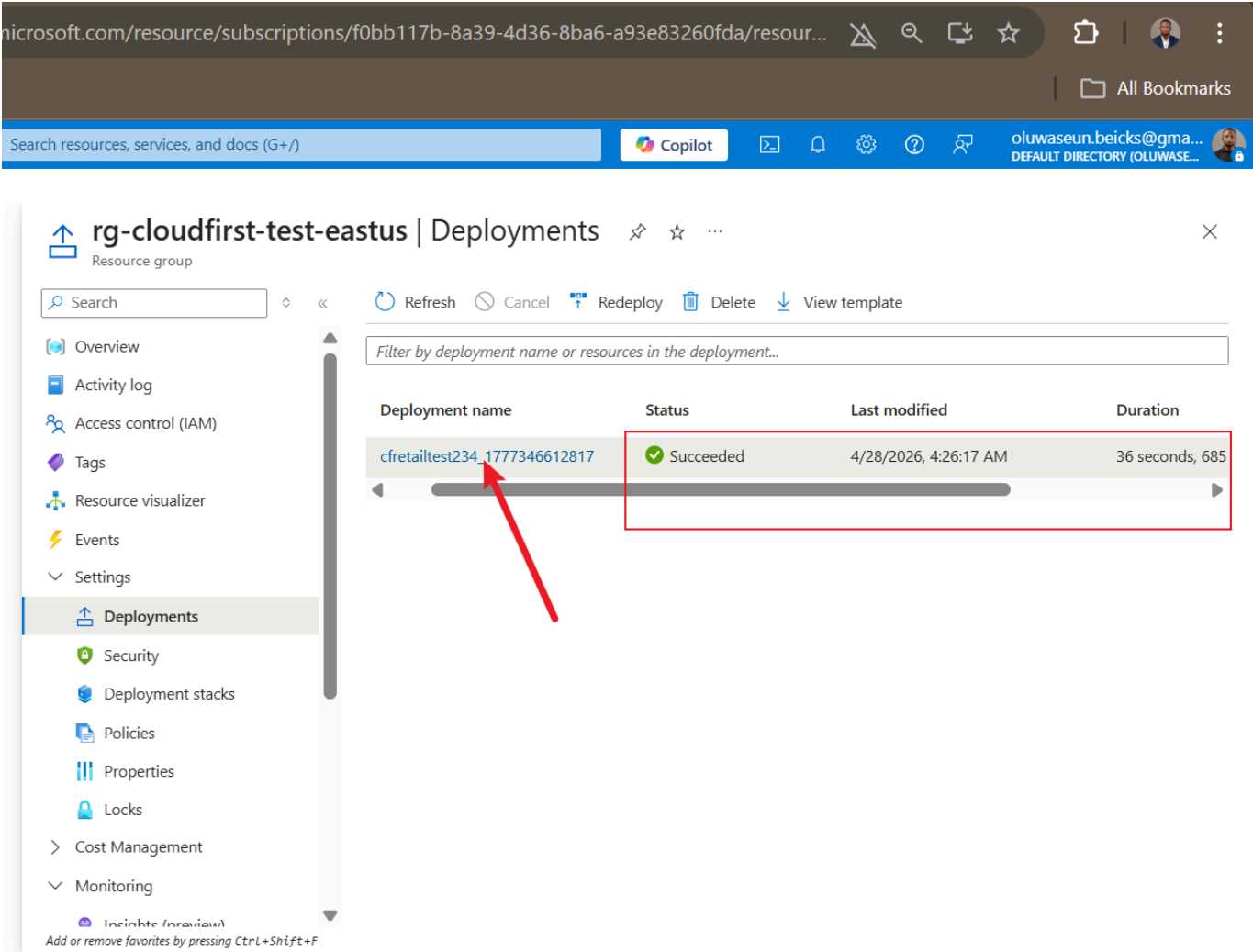
Business Impact: Without consistent resource-level tagging, a total cloud bill gives no visibility into which project or team is driving spend. With tags like `Project=CloudFirst-Retail` and `CostCenter=CC-1001`, finance can allocate costs precisely and identify overruns early.

Step 10 — View ARM Deployment History

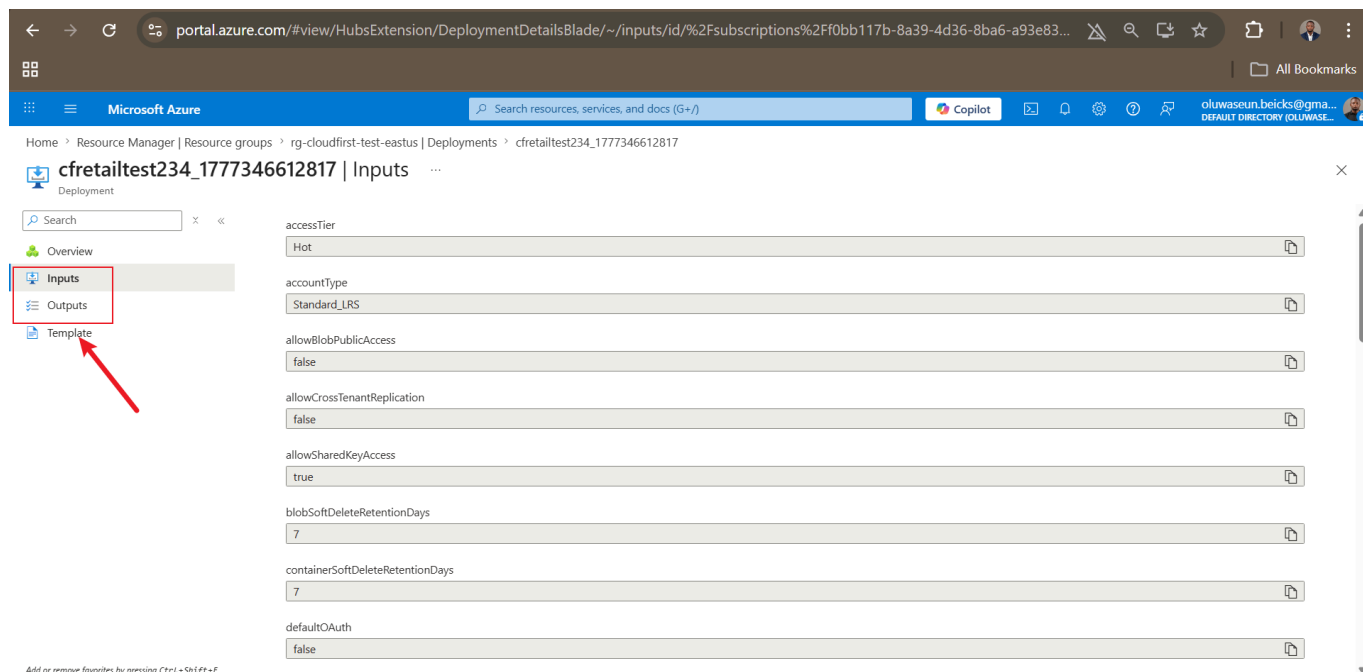
From the Resource Group, navigated to **Settings** → **Deployments** to view the Azure Resource Manager deployment audit trail.



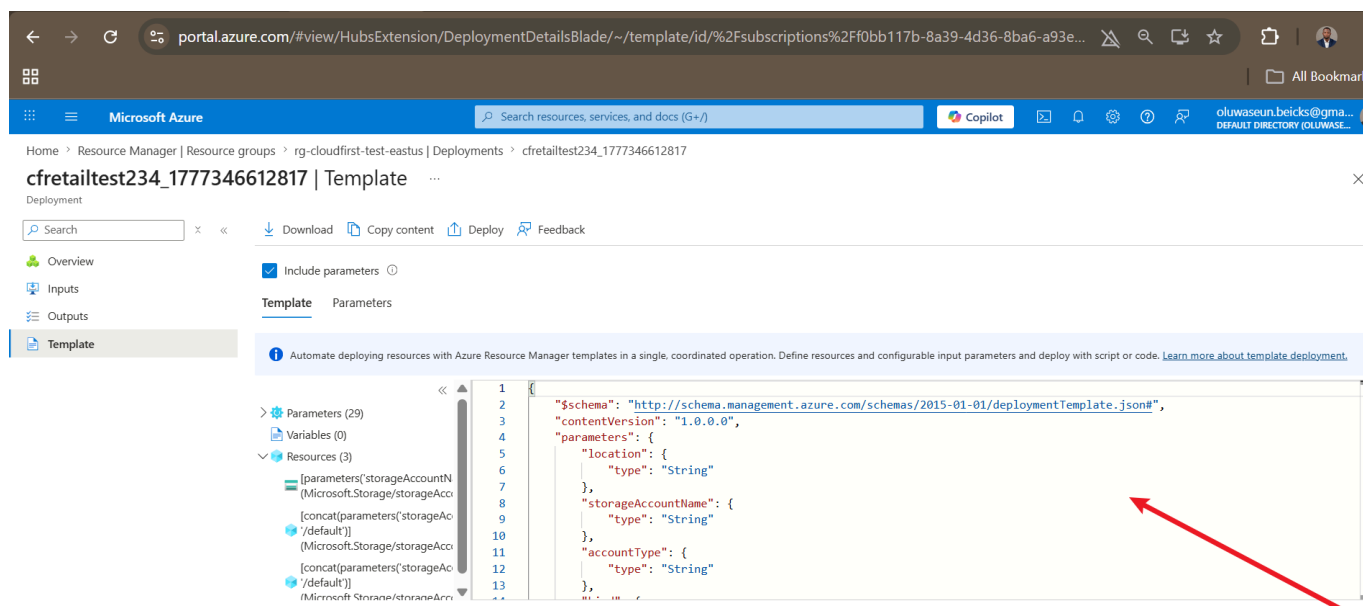
Reviewed the deployment list — noting the **Status** (Succeeded) and the **Duration** it took to provision the storage account.



Clicked on the deployment name and explored the **Inputs**, **Outputs**, and **Template** tabs.



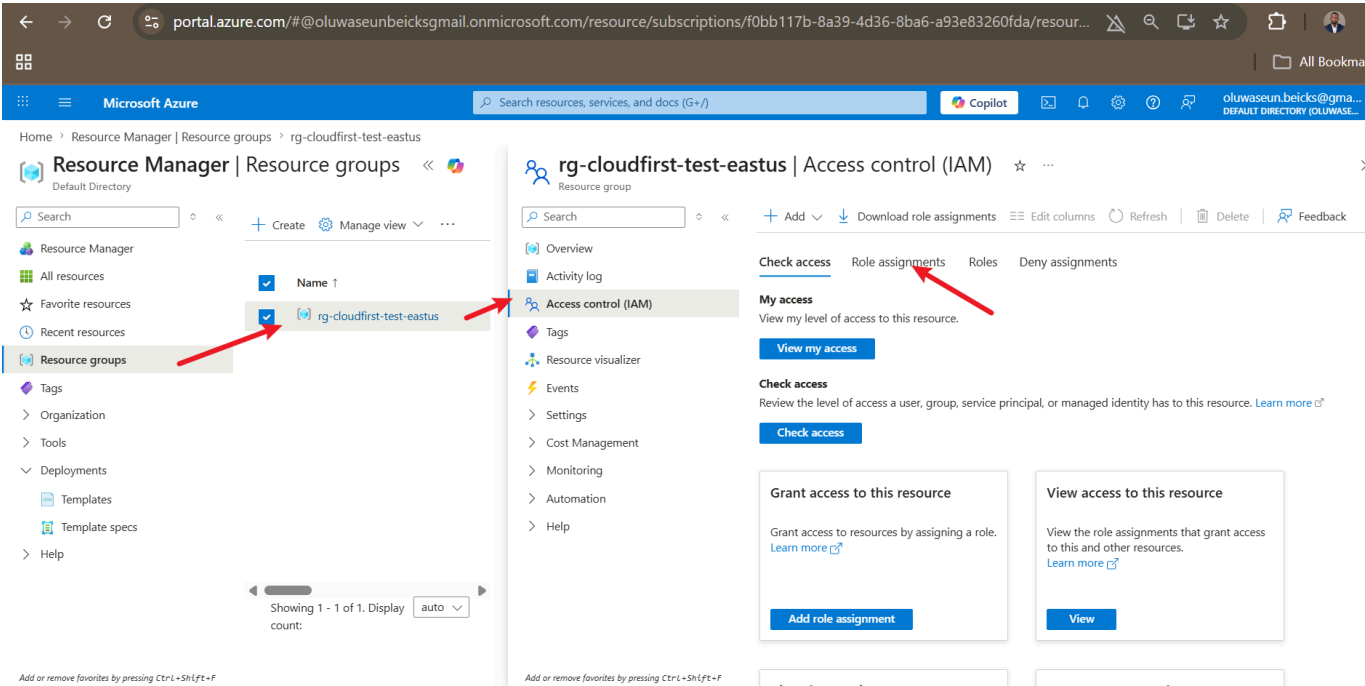
Viewed the underlying **ARM template** — the JSON infrastructure-as-code blueprint that Azure used to provision the storage account. This same template can be exported, version-controlled, and reused for repeatable deployments.



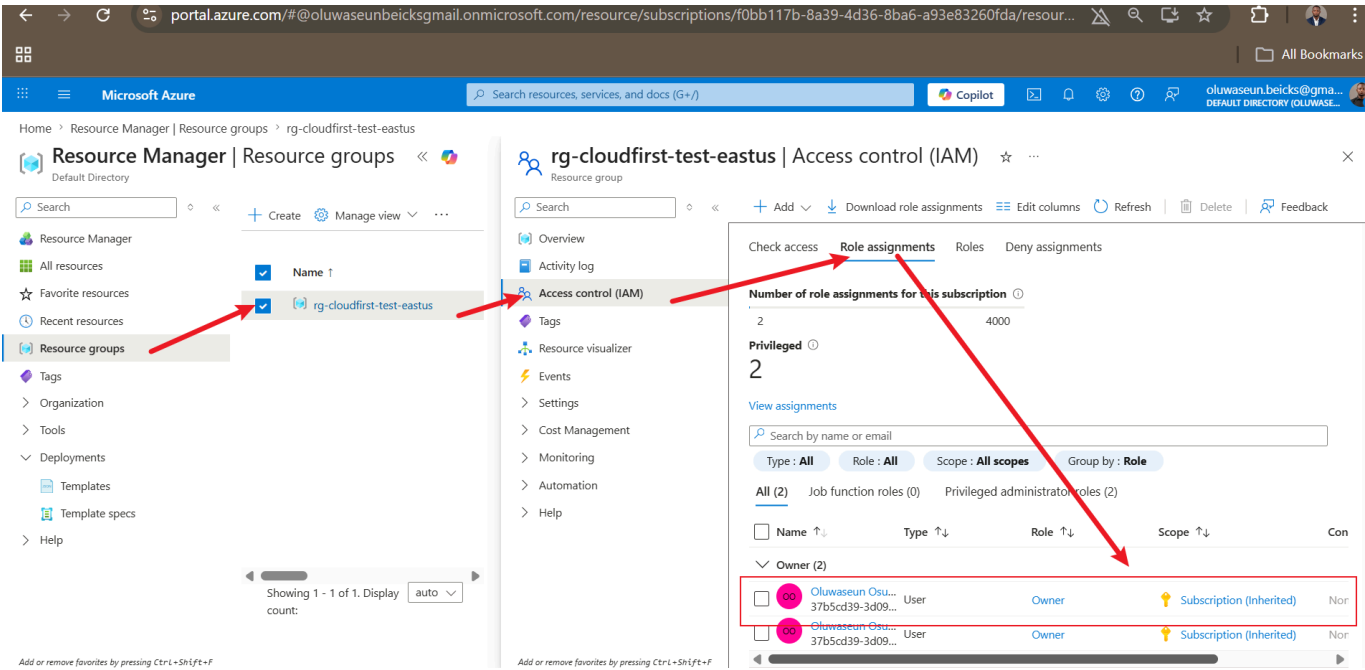
DevOps Connection: Every Azure deployment — whether made through the portal, CLI, or SDK — goes through ARM. The deployment history and exportable templates are the foundation of Infrastructure as Code (IaC) practices that senior cloud engineers rely on.

Step 11 — Explore RBAC and Policy Inheritance

From the Resource Group, navigated to **Access control (IAM)** → **Role assignments**.



Observed that my account has the **Owner** role — inherited from the Subscription level above. The **Scope** column confirmed the permission originated at the subscription, not the Resource Group itself.



Key Distinction — Inheritance Rules:	
Feature	Inherits Down?
RBAC Permissions	Yes
Azure Policies	Yes
Tags	No

Azure Governance Hierarchy (Visualised)


```
Root Management Group
├── Management Groups (up to 6 levels)
│   ├── Subscriptions (billing boundary)
│   │   ├── Resource Groups (logical container)
│   │   │   └── Resources (VMs, Storage, Databases...)
```

Permissions & Policies → flow DOWN 

Tags → do NOT flow down 

Key Takeaways

Resource Groups Are Logical, Not Physical Resource Groups only store metadata in their selected region. Resources inside can be deployed in any Azure region globally — demonstrated by deploying a storage account in South Africa North inside an East US Resource Group.

Tag Inheritance Is a Myth (The Most Common Exam Trap) Tags must be applied directly and individually to each resource. Tagging only the Resource Group gives a false sense of coverage and leads to incomplete cost reports in production environments.

RBAC and Policies DO Inherit Unlike tags, RBAC permissions and Azure Policies flow automatically from Subscription → Resource Group → Resource. This is the mechanism that enables governance at scale without manually configuring every resource.

ARM Powers Everything Every Azure operation — portal click, CLI command, SDK call — is processed through Azure Resource Manager. ARM's deployment history provides a full audit trail and the exported templates enable repeatable, version-controlled infrastructure deployments (IaC).

Region Selection Has Real Impacts Beyond geography, region choice affects latency, cost (up to 30–50% variance for the same service), data residency compliance, and service availability. Region selection is a deliberate architectural decision, not just a default.

Exam Tips (AZ-900)

Question Pattern	Answer
Can resources in a RG be in different regions?	Yes — RG is logical, not physical
What happens when you delete a Resource Group?	All resources inside are deleted permanently
Do tags on a RG flow to resources inside?	No — tags must be applied to each resource directly
What DOES inherit from RG to resources?	RBAC permissions and Azure Policies
What handles all Azure deployment requests?	Azure Resource Manager (ARM)

Question Pattern	Answer
What does ARM idempotence mean?	Running the same deployment multiple times produces the same result
What sits above Subscriptions in Azure governance?	Management Groups
How deep can Management Groups nest?	6 levels (excluding root)

Resources

- [Azure Portal](#)
- [Azure Resource Manager Documentation](#)
- [Azure Naming Conventions](#)
- [Dion Training — AZ-900 Course](#)

Lab completed by Oluwaseun Osunsola [[View LinkedIn](#)] as part of the Microsoft Azure Fundamentals (AZ-900) certification preparation — Section 3: Azure Architecture.